Technical Specifications for 6G Channel Knowledge Map Generation

High-Fidelity Ray-Tracing Requirements

TFG title: Channel Knowledge Map prediction with Deep Learning for 6G UAV Enabled Networks

Guillem Moreno Garcia

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1 Introduction

The objective of this TFG is to leverage Deep Learning to estimate **Channel Knowledge Maps (CKMs)** from 3D urban geometry. This project addresses the 6G requirement for "Environment-Aware Communication" and "Integrated Sensing and Communication" (ISAC).

To train a robust AI model, we require a dataset generated via deterministic Ray-Tracing (RT). The simulation must capture the specific propagation characteristics of 6G mmWave/Sub-THz frequencies, where geometry plays a causal role in signal distribution.

2 Deep Learning Context

The AI model (a Convolutional Neural Network) will operate as a "Physics-Informed Image Translator."

- **The Task:** The network maps a *Geometry Image* (Height Map) → *Radio Images* (Power, Delay, Angle).
- **The Challenge:** The network must implicitly learn Maxwell's equations. It must recognize that a building corner causes diffraction (signal bending) and a flat wall causes reflection (signal bouncing).
- **Requirement:** The input data must be high-resolution enough to capture these geometric features (corners, edges), and the output data must be physically accurate.

3 Simulation Configuration

3.1 Geometric Setup & Fidelity

The veracity of the "Digital Twin" is paramount. A standard 2D map is insufficient; we require a 2.5D Surface Model.

Environment: Deep Urban Canyon (Dense UMi)

Specification: A 500m × 500m area of a dense city center (e.g., Manhattan grid or European old town).

Rationale: 6G networks rely on "canyon propagation," where signals travel down streets by bouncing off walls. Open areas are too simple for the AI to demonstrate value.

Spatial Resolution: 1 **pixel** = 1 **meter**^{2}

Rationale: At mmWave frequencies, the transition from "Strong Signal" to "No Signal" (shadowing) happens over a few centimeters. A 1-meter resolution is the minimum required to capture the sharp shadow boundaries cast by buildings.

Format: Rasterized Digital Surface Model (DSM)

Specification: A matrix where $H_{x,y}$ = height of the structure in meters. Ground is 0.

Receiver (Rx) Grid

Specification: Uniform grid at 1.5m height (handset level).

3.2 Electromagnetic (EM) Configuration

3.2.1 Detailed Parameters

Table 1: Required Ray-Tracing Engine Parameters

Parameter	Value	Technical Justification
Frequency	6 - 24 GHz	Standard 6G FR3. Ensures valid reflection coefficients for concrete/glass.
Bandwidth	**100-800 MHz**	**Crucial for 6G.** High bandwidth provides high temporal resolution ($\Delta t \approx 1/B \approx 1.25$ns). This is necessary to resolve dense multipath clusters in the Delay Spread map, enabling ISAC sensing applications.
Tx Power	46 dBm	Standard Macro/Micro base station EIRP.
Antenna	Omni	(UAV) maximum upto 1km
Reflections	**Max 3**	**Waveguiding.** In 6G frequencies, signals cannot penetrate buildings. They must "bounce" down the street. We need at least 3 bounces to simulate coverage in Non-Line-of-Sight (NLoS) side streets.
Diffractions	Max 1	The buildings will be pure concrete, 20dBs of loss for 20GHz plus probably.

3.3 Line of Sight (LoS): The Physics and The Requirement

We explicitly require a Binary LoS Map output (1 if visible, 0 if blocked).

3.3.1 Why is this necessary?

In lower frequencies (4G/WiFi), LoS is just "one of many paths." (there is not as many loss) In 6G mmWave, LoS is a binary switch for the laws of physics. However it is true it's not calculated exactly the same way as Optical LoS, is not AS BINARY as it. So there is the binary aproximation and the augmented one. This is important for the regime, it can be rician or rayleigh:

1. **LoS Regime (Rician Fading):** The receiver sees the transmitter. The phase is stable.

2. **NLoS Regime (Rayleigh/Log-Normal):** The receiver is in a shadow. The signal relies entirely on scattering.

For the AI: By providing the LoS map as a target, we force the Neural Network to learn *Geometry Awareness*. The model learns to segment the city into "Visible Zones" and "Shadow Zones," applying different power prediction logic to each.

4 Required Dataset Ground Truth

The simulation must export pixel-perfect maps aligned with the input geometry.

4.1 Required Inputs

1. LoS matrix (binary)
2. Height matrix
3. Antenna binary matrix (where they are and their height)
4. Frequency (maybe?) (we'll train on 3 maybe, if only one no)
5. Bandwidth (maybe?)
6. TX Power (maybe?)

4.2 Required Outputs

Matrixes of (Tensor 6xMxN)

1. **Received Signal Strength (RSS) [dBm]:**
   - Total integrated power matrix.
   - *Purpose:* Coverage analysis and SNR heatmaps.
2. **Augmented Line-of-Sight (LoS) Map:**
   - Boolean visibility mask.
   - *Purpose:* Fundamental classification task for the AI; critical for ISAC positioning accuracy.
3. **RMS Delay Spread (σ_τ) [ns]:**
   - $\sigma_\tau = \sqrt{\frac{\sum P_i(\tau_i - \bar{\tau})^2}{\sum P_i}}$.
   - *Purpose:* Measures "Echo Richness." In 6G Sensing, this map reveals the distance to surrounding scatterers (walls).
4. **RMS Angular Spread (σ_θ) [deg]:**
   - Measures the directional variance of arriving energy.
   - *Purpose:* Determines the potential for Spatial Multiplexing (MIMO Capacity).
5. **Ricean K-Factor (K) [dB]**
   - $K = 10\log_{10}\left(\frac{P_{dominant}}{P_{scattered}}\right)$.
   - Purpose: Indicates channel reliability. High K-factor = stable link; Low K-factor = severe fast fading.

Ok maybe not K factor, its unrealistic because the wavelength at 16GHz is 1.7cm which means the fast fading would be very hard to calculate. DO NOT TAKE IT INTO ACCOUNT FOR THE REQUIREMENTS OF THE DATABASE.